Himalayan Climbing Expeditions

What predicts the probability of a successful summit attempt,
and does the effect of supplemental oxygen depend on peak height?

Steven Dahms · Bardaan Dhesi · Kritarsh Negi
STAT 3302 — Group Project

Research Questions & Motivations

Questions

- What factors predict the probability of a successful summit attempt?
- Is the relationship between peak height and success rate linear?
- Does the effect of supplemental oxygen depend on peak height?
  - Central hypothesis: oxygen helps more at extreme altitudes where hypoxia is severe

Motivations

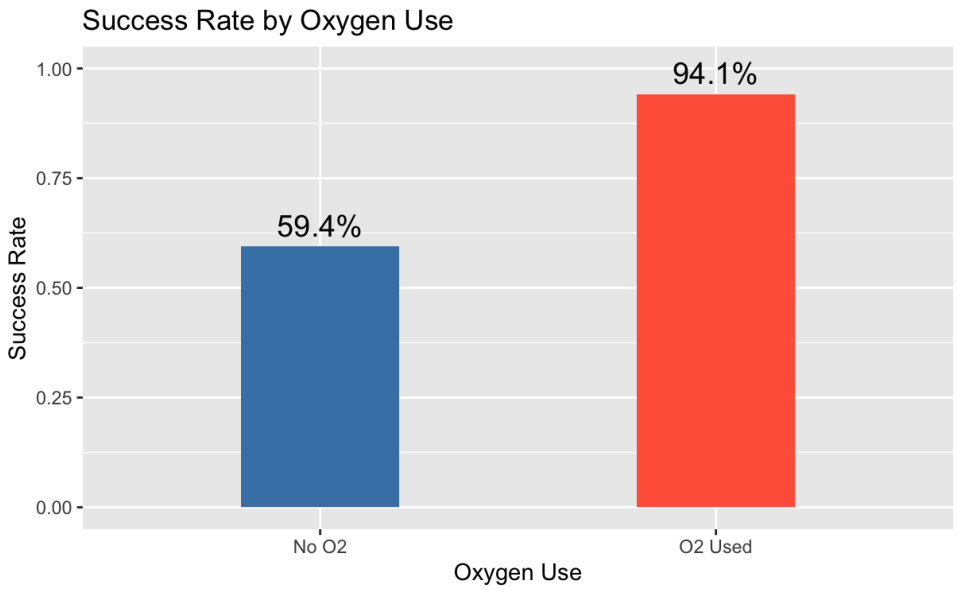
- Himalayan mountaineering carries serious risks
- Understanding what drives success can inform expedition planning
  - Interaction of oxygen and altitude is physiologically motivated, not just statistical
- Its plausible that climbers self-select onto harder peaks, meaning higher peaks attract more skilled and prepared teams, which suppresses the expected decline in success rate at seen at more moderate altitudes

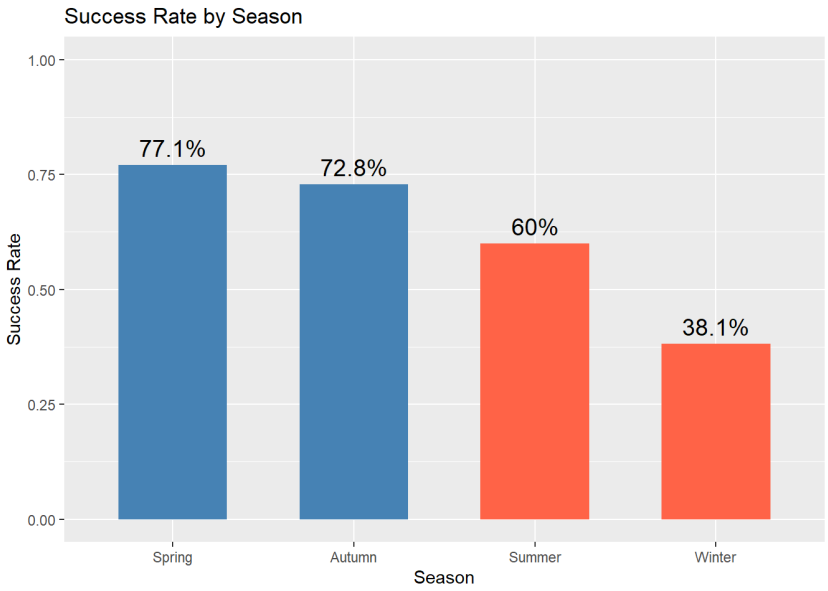
Data & Response Variable

n = 882	74.3%	490	5,407–8,849m
Expeditions (2020–2024)	Overall Success Rate	Himalayan Peaks	Height Range

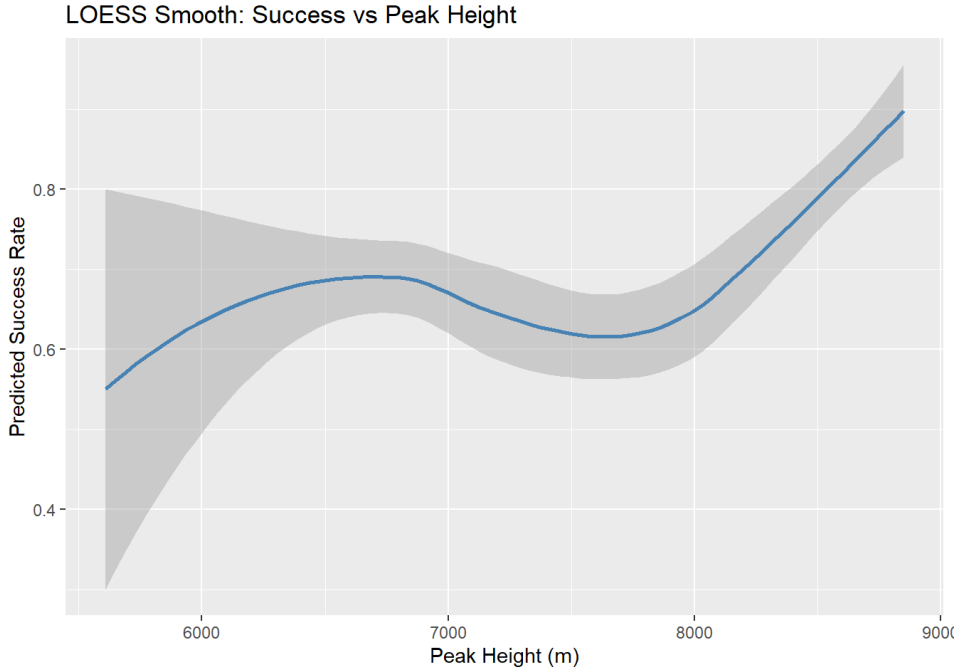
- *Unit of observation*: one full expedition attempt (success or failure)
- *Response variable:* SUCCESS = 1 if
  - TERMREASON ∈ {1, 2, 3 }, so includes subpeaks and claims
- *Cleaning and Organization*
  - 882 observations used after dropping records with missing peak height
  - Re-attempt are NOT treated as new observations (Ensuring Independence)

Exploratory Analysis





Exploratory Analysis Continued



- Success rate is non-monotonic across altitude
  - rises through mid-range peaks, dips until near 8000m, then increases sharply at extreme altitudes
- Counter-intuitive rise above 8000m may reflect self-selection: only elite, well-resourced teams attempt death-zone peaks (colloquially defined as above 8000 meters)

Model Specification

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \text{logit}(\pi_i) = \beta_0 + \beta_1 \cdot \text{O2USED} + \beta_2 \cdot \text{HEIGHTM_KM_C} + \beta_3 \cdot \text{HEIGHTM_KM_C}^2 + \beta_4 \cdot (\text{O2USED} \times \text{HEIGHTM_KM_C}) + \beta_5 \cdot \text{SEASON} + \beta_6 \cdot \text{CAMPS} + \beta_7 \cdot \text{TOTMEMBERS}$$

Predictor	Description	Rationale
O2USED	Supplemental oxygen used (T/F)	Directly compensates for altitude hypoxia
HEIGHTM_KM_C	Peak height, centered at mean (~7.76 km)	Taller peaks → thinner air, harder climb
HEIGHTM_KM_C²	Quadratic height term (centered)	Captures non-monotone altitude effect (self-selection at extreme peaks)
O2USED × HEIGHTM_KM_C	Interaction term	Central hypothesis: O2 benefit grows with altitude
SEASON_FACTOR	Season (Spring/Summer/Autumn/Winter)	Weather windows differ substantially by season
CAMPS	Number of high camps	More camps allow better acclimatization staging
TOTMEMBERS	Total team size	Larger teams share load, assist struggling members

Model Results — Odds Ratios

Predictor	Odds Ratio	95% CI	Significant?
O2USED (at mean height)	14.53	[6.34, 37.14]	✓ Yes — O2 helps at mean altitude
HEIGHTM_KM_C	0.088	[0.044, 0.165]	✓ Each extra km ~91% lower odds (no-O2)
HEIGHTM_KM_C²	0.529	[0.333, 0.828]	✓ Yes — curvature downward
CAMPS	1.765	[1.487, 2.104]	✓ Yes — more camps help
TOTMEMBERS	1.100	[1.057, 1.148]	✓ Yes — larger teams help
O2USED × HEIGHTM_KM_C	27.85	[7.49, 98.80]	**✓ YES — KEY FINDING**
SEASON: Spring	0.585	[0.368, 0.929]	✓ Yes — lower than Autumn
SEASON: Summer	0.972	[0.145, 8.029]	✗ No — crosses 1
SEASON: Winter	0.881	[0.278, 2.723]	✗ No — crosses 1

Baseline: AUTUMN. Height is centered at its mean (~7.76 km); main-effect ORs for O2USED and HEIGHTM_KM_C are interpreted at the mean altitude. Interaction OR is per km of altitude.

Interaction Term Justification

Likelihood Ratio Test

H_0: additive model vs H_1: with O_2 × height interaction

Δ deviance = 20.93 on 1 df
$p \approx 4.8 \times 10^{-6}$ ***

Adding the interaction significantly improves fit at $p < 0.001$. We reject the additive model.

AIC Comparison

Additive model: AIC = 674.58
Interaction model: AIC = 655.65

ΔAIC ≈ 18.93

Decision

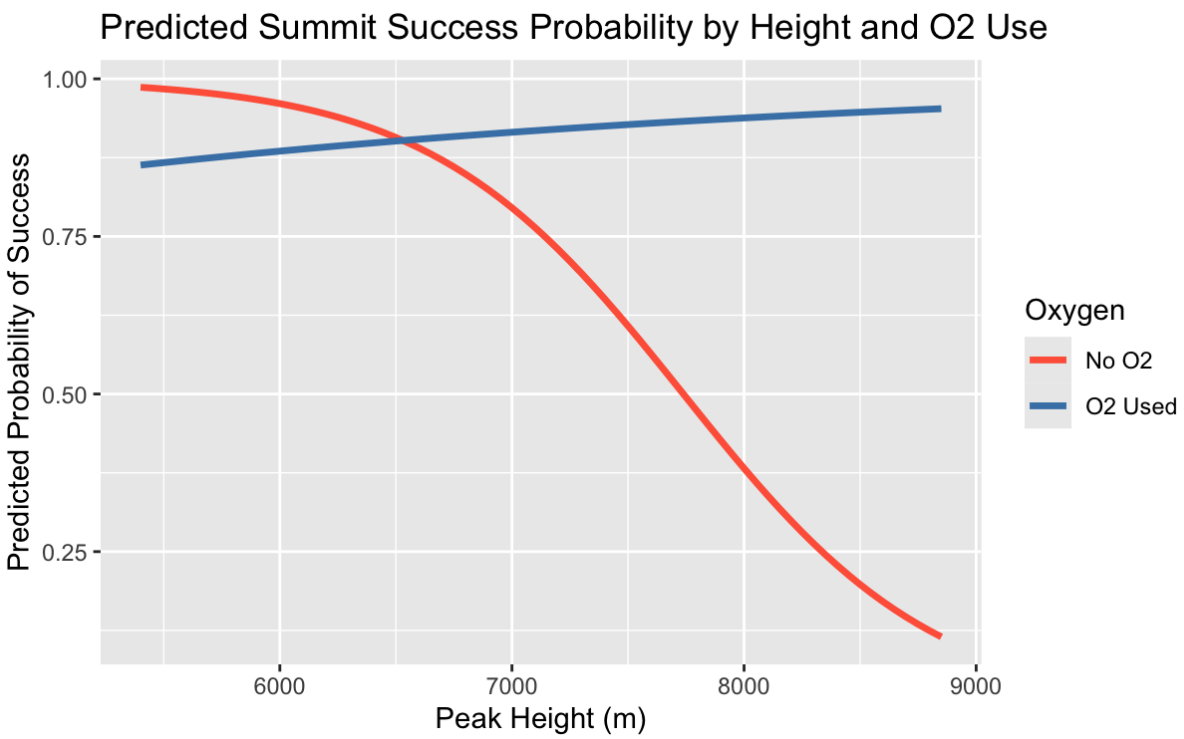
Retain the O2USED × HEIGHTM interaction.

Both criteria agree:

- LRT rejects additivity ($p \approx 5\times10^{-6}$)
- AIC prefers the interaction by ~18.9

Test run in R: `anova(model_additive, model, test = "Chisq")` and `AIC(model_additive, model)`

Key Finding — The Interaction Effect



- The interaction term is significant: OR = 27.85 [7.49, 98.80] per km of altitude
- At low altitudes, oxygen use provides little additional benefit
- At extreme altitudes, oxygen use dramatically increases predicted success probability
- CONFIRMS central hypothesis: ***the survival advantage of supplemental oxygen grows as hypoxia becomes more severe at altitude***

Model Diagnostics

ANOVA (Chi-Square)

Test confirms all retained predictors significantly reduce deviance ($p < 0.05$), validating model structure

Pairs Plot

Using ggpairs shows no severe collinearity between HEIGHTM_KM_C, CAMPS, TOTMEMBERS, and O2USED. Predictors are reasonably independent.

Model Comparison — Polynomial

Adding I(HEIGHTM_KM_C^2) significantly improves fit. AIC goes from 661 to 656 on $p = 0.005$.

Summary & Conclusions

- **Peak height significantly reduces success odds**
  - **taller peaks are harder**
- **More camps and larger teams significantly increase success odds**
- **The oxygen × height interaction is statistically significant (OR = 27.85 per km)**
  - Oxygen use matters more at extreme altitudes — confirming the physiological hypothesis
- **Spring expeditions have significantly lower success than Autumn; Summer and Winter do not differ significantly**

Limitations

- Small post-filter sample (n = 882) limits power for complex interactions
- TERMREASON codes 2 & 3 (subpeak / claimed) are very rare — may affect robustness
- Causal interpretation not possible since its observational data only